Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials

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September 07, 2026

Abstract

Background. Trial designers who plan a longitudinal study with a fixed total observation budget face an allocation question. How should observations be distributed across run-in (J_0), treatment (J_1), and common close (J_2) phases to minimize the variance of the treatment-effect estimator? Existing guidance is heuristic, and the optimum has not been characterized for the three-phase run-in setting. **Methods.** We formulate the allocation question as a discrete optimization problem over the Woodbury-based variance formula derived in Paper 1 of this compendium. We solve it by exhaustive enumeration of all feasible phase splits. We describe the behavior of the optimum as a function of the signal-to-noise ratio σ_b/σ. Extensions cover replicated endpoint observations (closely spaced measurements at the end of treatment that average out measurement error on the final assessment, analogous to run-in averaging applied at the post-treatment end), and pattern-mixture handling of monotone dropout following the Frost-Kenward-Fox (2008) framework. **Results.** We find that under a fixed total budget, the optimal design places zero observations in the run-in phase in every configuration examined. The budget instead splits between the treatment and common close phases. This finding reconciles with the run-in efficiency gains of Paper 1. That is, run-in observations help when they are additional visits appended to a fixed treatment phase. However, when visits are fungible, the run-in phase is never chosen, because it introduces the pre-randomization slope parameter without contributing to the treatment contrast. The common close phase enters the optimal design once σ_b/σ exceeds a budget-dependent crossover, which falls from 0.81 at $J = 3$ to 0.30 at $J = 8$. At high ratios, roughly half the budget goes to the common close. Replicating the final endpoint reduces the treatment-effect variance by up to 31% at $\sigma_b/\sigma = 0.5$ (five replicates), but by at most 5% at $\sigma_b/\sigma = 2$. **Conclusions.** These numerical allocation results give trial designers a simple decision procedure driven by the signal-to-noise ratio and the visit budget. The procedure extends to replicated-endpoint and dropout-adjusted designs within the same variance framework.

Contents

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1 Introduction

We begin with the relative efficiency surface in Paper 1 (Figure 1), which shows diminishing returns from additional run-in observations appended to a fixed treatment phase. Most of the attainable variance reduction is recovered by the first two run-in observations, and the marginal value falls off quickly thereafter; Section 3 examines this marginal pattern, including a penalty specific to a single run-in observation. Common close observations improve efficiency in a similar way, but at a rate that depends on the number of treatment observations J_1.

These observations raise a design question that Paper 1 does not address. Given operational constraints on the total number of visits or the total trial duration, what is the optimal distribution of observations across the three phases? That is the question we take up here.

[1] calculated optimal time-point positions for linearly divergent treatment effects under compound symmetry, but did not consider the run-in setting.[2] derived sample sizes for trends across repeated measurements under missing data.[3] showed that having more than one pre-treatment measurement is beneficial for designs analyzed via ANCOVA.[4] provided sample size formulas for repeated measurement designs.[5] developed general guidelines for the design of clinical trials with longitudinal outcomes.

We note that the question posed here is a design-optimization problem and belongs to a well-developed tradition. The general theory of optimum experimental design, in which a design is chosen to optimize a functional of the Fisher information matrix, is set out by[6] and[7].[8] extended the theory to random-effects regression models, the model class used here, giving an algorithm that maximizes the determinant of the population information matrix. Closest in setting is[9], who constructed D-optimal cohort designs for linear mixed-effects models with a cost function for repeated measurements, and found that concentrating measurements at optimally chosen time points in a single cohort is efficient. Three features distinguish the present problem from that literature. First, the design space is not a set of time points to be placed freely but a partition of a fixed count of equally spaced visits into three phases with a prescribed ordering. Second, the run-in phase introduces its own nuisance parameter, the pre-randomization slope δ, so enlarging that phase both adds information and consumes a degree of freedom; the optimal-design literature usually holds the parameter vector fixed as the design varies. Third, the criterion here is the variance of a single scalar contrast, a c-optimality criterion, rather than the D-optimality that most of this literature adopts. The results below are accordingly numerical rather than an application of the general equivalence theory.

The slopepower command[10] and the longpower R package[11] implement power calculations for slope-comparison trials but do not optimize the allocation of observations across phases.

2 Problem formulation

2.1 Fixed total visits

Given J total observations (excluding baseline), choose (J_0^*, J_1^*, J_2^*) to minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = J$, $J_0 \geq 0$, $J_1 \geq 1$, $J_2 \geq 0$.

The objective function is $V(J_0, J_1, J_2) = [(X'\Sigma^{-1}X)^{-1}]_{\gamma\gamma}$ computed via the Woodbury identity from Paper 1.

Throughout, the phase counts (J_0, J_1, J_2) correspond to the arguments (r, k, f) of the `runinpower` package routines used in the code chunks and tables.

2.2 Fixed total duration

Alternatively, fix the trial duration $T = (J_0 + J_1 + J_2) \cdot t$ and minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = T/t$, or more generally allow non-equally-spaced observations within each phase and minimize over the observation times as well.

Table 1: Marginal variance reduction from the J_0-th run-in observation ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	ΔV_{J_0}
1	0.1765
2	0.6667
3	0.5124
4	0.3176
5	0.2008
6	0.1342
7	0.0945
8	0.0696

Table 2: Boundary of the $J_0 = 1$ anomaly ($J_1 = 2$, $J_2 = 0$, $\sigma = t = 1$). The reversal $\Delta V_1 < \Delta V_2$ holds only between the two computed boundary ratios 0.53 and 1.51.

σ_b/σ	ΔV_1	ΔV_2	$\Delta V_1 < \Delta V_2$
0.32	0.1633	0.0080	no
0.53	0.0454	0.0418	no
0.75	0.0033	0.2653	yes
1.00	0.1765	0.6667	yes
1.25	0.6649	1.1082	yes
1.51	1.5407	1.5352	no
2.00	4.2000	2.1718	no

3 Diminishing returns from run-in observations

3.1 Marginal value of the J_0-th run-in observation

Define the marginal variance reduction from adding the J_0-th run-in observation:

$$\Delta V_{J_0} = V(J_0 - 1, J_1, J_2) - V(J_0, J_1, J_2) \tag{1}$$

3.2 The $J_0 = 1$ anomaly

A non-monotonicity is visible in the table: we see that $\Delta V_1 < \Delta V_2$ at the displayed parameter setting ($\sigma_b/\sigma = 1$). That is, the first run-in observation is less valuable than the second. This occurs because adding a single run-in observation forces the model from 2 parameters (β, γ) to 3 parameters (δ, β, γ), consuming a degree of freedom. The second run-in observation does not increase the parameter count and so provides pure information gain. The reversal is not universal, however. A numerical scan (Table 2) shows that, for $J_1 = 2$, $J_2 = 0$, and $\sigma = t = 1$, the inequality $\Delta V_1 < \Delta V_2$ holds only for σ_b/σ in an intermediate band, roughly $(0.53, 1.51)$. At lower and higher ratios the first run-in observation is the more valuable one.

This effect is loosely analogous to[3]'s finding that a single pre-treatment measurement used as

Table 3: Optimal allocation (J_0^*, J_1^*, J_2^*) for fixed total observations J, by signal-to-noise ratio.

J	Low ($\sigma_b/\sigma = 0.32$)	Med ($\sigma_b/\sigma = 1.0$)	High ($\sigma_b/\sigma = 2.24$)	AD ($\sigma_b/\sigma = 1.70$)
3	(0,3,0)	(0,2,1)	(0,2,1)	(0,2,1)
4	(0,4,0)	(0,2,2)	(0,2,2)	(0,2,2)
5	(0,5,0)	(0,3,2)	(0,3,2)	(0,3,2)
6	(0,6,0)	(0,3,3)	(0,3,3)	(0,3,3)
7	(0,5,2)	(0,4,3)	(0,4,3)	(0,4,3)
8	(0,5,3)	(0,4,4)	(0,4,4)	(0,4,4)

an ANCOVA covariate can reduce power when the pre-post correlation is low. The practical implication is regime-specific: **within the intermediate signal-to-noise band of Table 2, a trial designer who adds run-in observations should plan at least two, not one.** Outside that band, the first run-in observation carries the larger marginal value, and no special caution applies.

3.3 Correlation structure governs diminishing returns

The correlation between the J_0-th run-in change score and the treatment-period change scores determines the information gain. Under compound symmetry:

$$\mathrm{Corr}(C_{-J_0}, C_l) = \frac{\sigma_b^2 t^2}{2\sigma^2 + \sigma_b^2 t^2} \tag{2}$$

which is constant in J_0. We note that the diminishing returns for $J_0 \geq 2$ arise from the information matrix structure, not from declining correlation. Under AR(1)[12], on the other hand, the correlation does decline with temporal distance, producing faster diminishing returns from distant run-in observations.

4 Optimal allocation for fixed J

The dominant feature of Table 3 is that the optimal run-in count is zero in every cell. That is, across all budgets $J = 3, \ldots, 8$ and all four signal-to-noise settings, including the Alzheimer's neuroimaging regime, the optimizer splits the budget between the treatment and common close phases and never assigns a visit to the run-in phase. The mechanism is a degree-of-freedom cost. Any positive run-in count forces the model to estimate the pre-randomization slope δ in addition to (β, γ), and run-in observations inform the treatment contrast only indirectly, through the random slope. A visit placed in the treatment or common close phase buys the same slope information without the extra parameter. This finding does not contradict the efficiency gains of Paper 1. There, run-in observations are additional visits appended to a design whose treatment phase is held fixed, and the comparison is against the shorter design without them. Here, visits are fungible under a fixed total budget, and a run-in visit always has a better use elsewhere. The two framings answer different design questions, and that distinction is the central practical message of this paper.

Table 4: Signal-to-noise ratio above which reallocating the last budgeted visit from the treatment phase to the common close, $(0, J-1, 1)$ versus $(0, J, 0)$, reduces $\mathrm{Var}(\hat{\gamma})$.

J	Crossover σ_b/σ
3	0.806
4	0.592
5	0.471
6	0.392
7	0.336
8	0.295

5 When is the common close phase beneficial?

The common close phase contributes information about the random slope b_i without contributing to the treatment comparison directly. When a common close observation is added to an otherwise fixed design, it reduced $\mathrm{Var}(\hat{\gamma})$ in every configuration we examined ($J_1 = 1, \ldots, 4$ with two run-in observations, and σ_b/σ down to 10^{-3}). That is, a free observation is always at least weakly helpful. The design-relevant question is therefore not whether the common close helps in isolation, but whether a visit taken from a fixed budget is better spent on the common close than on the treatment phase. We quantify that trade-off next.

5.1 Crossover as a function of the budget J

Table 4 shows the crossover ratio at which the reallocation of a single budgeted visit from the treatment phase to the common close becomes favorable. The crossover falls monotonically with the budget, from $\sigma_b/\sigma \approx 0.81$ at $J = 3$ to ≈ 0.30 at $J = 8$. That is, larger trials should commit visits to the common close at progressively lower signal-to-noise ratios. Above the crossover, the full optimizer (Table 3) assigns an increasing share of the budget to the close phase. At $J = 8$ the optimum is an even split $(0, 4, 4)$ at every ratio examined from 0.6 upward, and even at $\sigma_b/\sigma = 0.32$ the optimum is $(0, 5, 3)$. These values are numerical results from exhaustive enumeration; an analytical characterization of the crossover remains open.

6 Non-equally-spaced designs

6.1 General time points

We note that the display in this section comes from a Mathematica notebook derivation for a *different* model than the one used elsewhere in this paper: a raw-outcome random-slope-only model with centered observation times ($\sum t_i = 0$) and independent residuals $R = \sigma^2 I$, rather than the change-score model with shared baseline error $R = \sigma^2(I + \mathbf{11}')$ that underlies all preceding sections. Under that raw-outcome model, with the normalization $\sum t_i^2 = 1$, the notebook gives for two time points t_1, t_2:

$$\mathrm{Var}(\hat{\gamma}) = \frac{3\sigma^2}{1 - t_1 t_2} + 2\sigma_b^2 \tag{3}$$

This expression has not been re-derived or numerically verified within the change-score framework of this paper, and reconciling the two formulations remains open.

Within the raw-outcome model above, the expression is minimized when $t_1 t_2$ is minimized (most negative), that is, when the two time points are maximally separated. This is consistent with the known result that observations at the extremes of the follow-up period are optimal for slope estimation. Whether and how this carries over to the change-score model of the preceding sections has not been established here.

Scope. The general (J_0, J_1, J_2) optimization in which observation times within each phase are also free parameters, rather than fixed and equally spaced, is not addressed in this paper. It connects to the optimal time-point positioning literature (e.g., Winkens et al. 2005) and we leave it as future work rather than develop it here.

7 Replicated endpoint observations

7.1 Motivation

The run-in mechanism averages multiple pre-randomization observations to reduce the contribution of measurement error σ^2 to the estimated baseline slope[3,13].[14] showed that replicated measures increase power, but with diminishing returns past 20-50 replications. The same logic applies at the end of the treatment period. If the final scheduled visit is replaced by (or augmented with) two or three closely spaced measurements, all under the same treatment condition, the measurement error on the endpoint assessment is reduced by averaging.

Concretely, if the last visit at time $J_1 t$ is replicated n_{rep} times over a short interval (days to weeks, negligible relative to t), the effective residual variance for the averaged endpoint becomes σ^2/n_{rep}, while the random slope contribution $\sigma_b^2 t^2$ is essentially unchanged. Similarly, the baseline visit at $j = 0$ can be replicated to reduce σ^2 in the change-score covariance.

We note that this is distinct from the common close design of Section 5. Common close observations are at *new* time points after treatment stops, providing information about the off-treatment slope. Replicated endpoints, on the other hand, are at the *same* time point as the last treatment visit, providing only measurement-error reduction.

7.2 Variance formula with replication

The residual covariance matrix R is modified so that the diagonal element corresponding to a replicated visit uses σ^2/n_{rep} instead of σ^2, and the off-diagonal elements reflecting the shared baseline error use $\sigma^2/n_{\text{rep,pre}}$. The Woodbury computation then proceeds identically.

The table confirms that endpoint replication is most beneficial when measurement error dominates (σ_b/σ small). When between-subject slope variability is large ($\sigma_b/\sigma > 2$), the gain from replication is modest, because the variance is then dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

Table 5: Fractional variance reduction from replicating the final endpoint ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$).

n_{rep}	$\sigma_b/\sigma = 0.5$	$\sigma_b/\sigma = 1$	$\sigma_b/\sigma = 2$	$\sigma_b/\sigma = 5$
1	0.000	0.000	0.000	0.000
2	0.182	0.091	0.030	0.005
3	0.250	0.125	0.042	0.007
5	0.308	0.154	0.051	0.009

7.3 Replication vs additional run-in observations

A natural question is whether a fixed budget of additional measurements is better spent on run-in observations, at distinct time points separated by t, or on endpoint replicates, which are closely spaced and offer measurement-error reduction only.

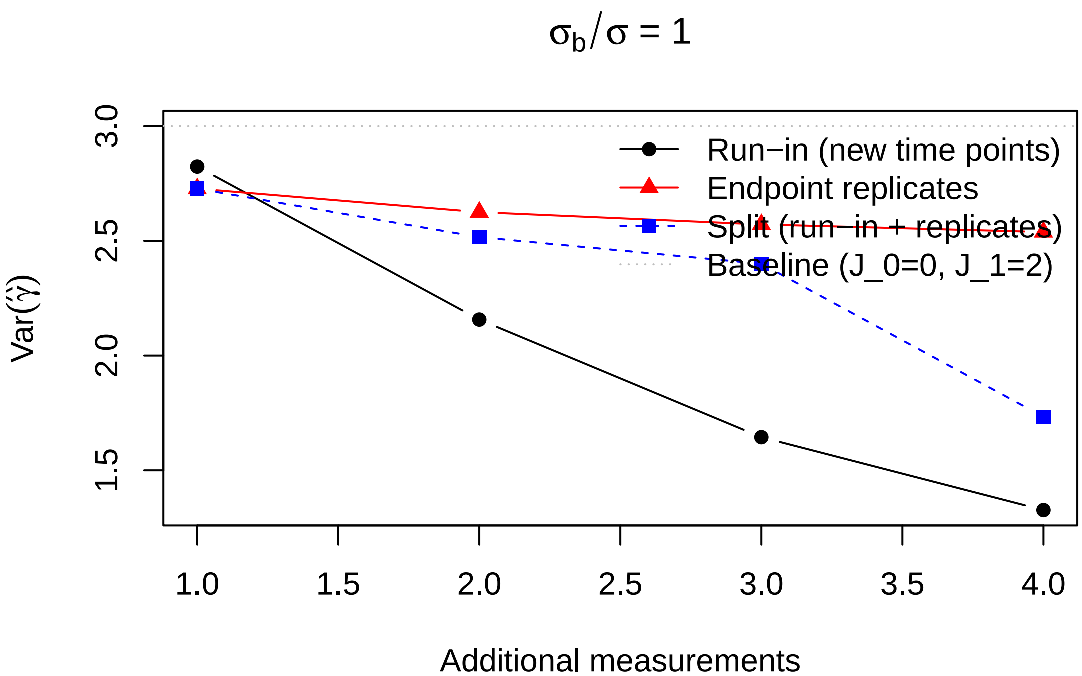

Figure 1: (#fig:rep_vs_runin)Variance as a function of additional measurements allocated either to run-in (new time points) or endpoint replication (same time point). Baseline: $J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$.

At $\sigma_b/\sigma = 1$, the setting of the figure, the comparison depends on how many extra measurements are available. With a single extra measurement, the endpoint replicate is the better use (variance 2.73 versus 2.82 for one run-in observation), because the lone run-in observation pays the degree-of-freedom penalty of Section 3. From the second extra measurement onward, however, run-in observations at new time points win (2.16 versus 2.63 at two extras, 1.64 versus 2.57 at three), because they provide information about the individual slope b_i, not just measurement-error reduction. When $\sigma_b/\sigma < 1$, endpoint replication is stronger still: at $\sigma_b/\sigma = 0.5$ it beat run-in at every budget we examined (one to four extra measurements), since measurement error is then the dominant source of variance.

Table 6: Per-pair variance with baseline and endpoint replication ($J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

n_{pre}	$n_{\text{post}} = 1$	$n_{\text{post}} = 2$	$n_{\text{post}} = 3$
1	3.0000	2.7273	2.6250
2	2.7273	2.5000	2.4138
3	2.6250	2.4138	2.3333

Table 7: Sample size adjusted for dropout via pattern-mixture ($J_0 = 2$, $J_1 = 2$, $J_2 = 1$, MIRIAD parameters).

Annual dropout	N (no dropout)	N (with dropout)
0.0	104	104
0.1	104	128
0.2	104	160
0.3	104	204

7.4 Combined replication at both ends

Replicating both the baseline and endpoint visits provides symmetric measurement-error reduction, as we show next.

8 Incorporating dropout

8.1 Pattern-mixture approach

Following[13] (Section 2.3), we handle dropout via a pattern-mixture model[15,16]. Participants are stratified by their observed visit pattern, and the overall variance is a weighted average:

$$N = \frac{1}{\sum_s p_s / N_s} \tag{4}$$

where p_s is the proportion with pattern s and N_s is the sample size that would be needed if all participants had pattern s. Under monotone dropout, the pattern for a participant who leaves after post-baseline visit j is the truncated design that retains the run-in and baseline visits plus the first j post-baseline observations. A participant with no post-baseline visit contributes no information about the treatment contrast, so that stratum enters with $N_s = \infty$. We ignore dropout during the pre-randomization run-in here, since run-in dropouts are typically replaced before randomization. This approximation treats each stratum independently.[17] derive exact variance formulas under monotone dropout for the three-component random coefficient regression model ($\sigma^2, \sigma_a^2, \sigma_b^2$), accounting for the effect of dropout on the estimability of the random slope variance. Their approach is more rigorous than the pattern-mixture approximation used here, which is conservative when dropout is informative about the random slope.

Table 8: Top 10 allocations for $J = 5$ observations (MIRIAD parameters, $\Delta = 0.25$, 90% power).

(J_0, J_1, J_2)	$\mathrm{Var}_1(\hat{\gamma})$	N per group
(0,3,2)	0.291783	50
(0,2,3)	0.303230	51
(1,2,2)	0.453050	77
(2,3,0)	0.551570	93
(0,4,1)	0.560024	95
(1,3,1)	0.587888	99
(3,2,0)	0.593958	100
(2,2,1)	0.616148	104
(0,1,4)	0.701145	118
(1,1,3)	0.803845	136

9 Numerical examples

9.1 AD neuroimaging trial

Using MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $t = 1$ year), we compare the optimal allocation for $J = 4, 5, 6$ total observations with and without dropout.

10 Discussion

The central allocation result is negative and, we believe, useful. Under a fixed total visit budget, the optimal design places no observations in the run-in phase in any configuration examined, including the Alzheimer's neuroimaging regime (Table 3). The budget instead splits between the treatment and common close phases. The mechanism is the degree-of-freedom cost identified in Section 3: any positive run-in count adds the pre-randomization slope parameter δ, while run-in observations inform the treatment contrast only indirectly. This does not contradict the efficiency gains reported for run-in designs in Paper 1 and the prior literature; those gains arise when run-in visits are additional measurements appended to a fixed treatment phase. When visits are fungible, as under a per-participant visit budget, the same visit buys more information in the treatment or common close phase. Trial designers should therefore ask which question their operational constraint poses: extra visits, where run-in can help, or reallocated visits, where run-in never helps in the settings examined here.

The common close allocation is governed jointly by the signal-to-noise ratio and the budget. A free common close observation always helped in the configurations we examined (Section 5), so the operative question is the budgeted trade-off: moving one visit from the treatment phase to the common close reduces the variance once σ_b/σ exceeds a crossover that falls with the budget, from about 0.81 at $J = 3$ to 0.30 at $J = 8$ (Table 4). Above the crossover, the optimizer assigns the close phase a growing share of the budget, up to an even split at $J = 8$ in the settings examined. In practice, high slope-to-noise outcomes such as neuroimaging endpoints sit well above the crossover at realistic budgets, so a substantial common close phase is warranted there. Low slope-to-noise outcomes, on the other hand, justify a common close only at larger budgets. Estimating σ_b/σ from pilot data is therefore the prerequisite design step.

Two refinements operate within a fixed phase allocation. First, when observation times within a phase are free, the two-point raw-outcome special case of Section 6 indicates that spacing observations toward the extremes of the window minimizes the slope-estimation variance; the corresponding optimization in the change-score model of this paper remains open. Second, replicating the baseline or endpoint visit reduces the measurement-error contribution to the change score without adding new time points. This is most valuable at low slope-to-noise ratios (up to a 31 percent variance reduction at $\sigma_b/\sigma = 0.5$ with five replicates) and offers little when slope variability dominates (at most 5 percent at $\sigma_b/\sigma = 2$). The comparison of Section 7 clarifies the choice between spending an additional measurement on a new run-in time point versus an endpoint replicate. At $\sigma_b/\sigma = 1$ the replicate wins when only one extra measurement is available, because a lone run-in observation pays the degree-of-freedom penalty, while run-in observations win from the second extra measurement onward by contributing slope information. Below $\sigma_b/\sigma = 1$, replication won at every budget we examined.

Several assumptions bound the scope of these recommendations. The allocation optima are computed under equally spaced observations, linear individual trajectories, and a compound-symmetric residual structure; the AR(1) analysis of Paper 2 implies that under decaying correlation the optimum shifts toward fewer and more closely spaced run-in observations, since distant ones lose informativeness. The allocation results themselves are numerical, obtained by exhaustive enumeration of the feasible phase splits; an analytical characterization of the optimum, for example by signing the marginal value of a close-phase visit, remains open. Dropout is accommodated through the pattern-mixture approximation of Section 8, applied to monotone truncation strata, which the exact monotone-dropout treatment of[17] would refine. Extending the allocation optimization jointly over phase counts, within-phase timing, and the correlation structure remains, in our view, the natural next step.

11 Conflict of interest

The author declares no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The `runinpower` R package implementing the optimal-allocation routines (`allocation.R` and `replicated.R`, with the design matrices they rely on built in `design.R`), the validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository at https://github.com/rgt47/runinpower. We list specific paths in the Reproducibility section below.

14 Reproducibility

We rendered this manuscript with R 4.4.x. The principal packages used are the in-package `runinpower` source, `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. All results in this manuscript are deterministic evaluations of the variance formulas given fixed input parameters. No Monte Carlo simulation is used, and no random seed is required.

Data and code availability. The R package source is at `R/`. The companion manuscripts (Papers 1, 2, 4) are at `analysis/report/{01,02,04}-*/`. The manuscript source and shared bibliography (`references.bib`, a symlink to `../../references.bib`) are at `analysis/report/03-allocation/`.

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